Insurance Portfolio Data Simulation & Analytics Report

---

1. Overview

This project simulates an automobile insurance portfolio to evaluate **premium revenue, claims behavior, and portfolio risk exposure**. Using PostgreSQL, a dataset of **1,000,000 insurance policies sold in 2024** was generated and analyzed based on predefined business rules for policy tenure, premium pricing, and claim occurrences.
The analysis examines key metrics such as **premium revenue, claim costs, claim ratios, and future claim liabilities**. The results were further visualized using an **interactive Power BI dashboard**, enabling exploration of claim trends, tenure performance, and portfolio profitability.
The findings highlight the relationship between **policy tenure, claim exposure, and long-term financial risk**, providing insights that can support pricing strategy and risk management decisions.

2. Data Simulation Methodology

Policy Sales Data Generation

A synthetic dataset representing **1,000,000 vehicle insurance policies sold in 2024** was generated.
Key steps followed:
• Policy purchases were **distributed evenly across all days of 2024**
• Each policy was assigned a unique **Customer_ID and Vehicle_ID**
• Policy tenure distribution was applied according to business rules

Policy Tenure	Distribution
1 Year	20%

2 Years	30%
3 Years	40%
4 Years	10%

Policy attributes were calculated as follows:

Vehicle Value: ₹100,000

Premium: ₹100 per year of policy tenure

Policy Start Date: Purchase Date + 365 days

Policy End Date: Start Date + Policy Tenure

This approach replicates the lifecycle of a real insurance policy and ensures consistency across the dataset.

Claims Data Simulation

Claims were generated based on predefined scenarios.

Claims in 2025

• Vehicles purchased on the **7th, 14th, 21st, and 28th** of each month were eligible

• **30% of eligible vehicles filed a claim**

• Claim date was set as **policy start date**

• Each vehicle could file **only one claim per year**

Claims in 2026

An additional claim event was simulated for:

• **10% of policies with 4-year tenure**

• Claim period: **Jan 1 – Feb 28, 2026**

• Claims distributed across **59 days**

Claim Amount:

10% of vehicle value = **₹10,000**

3. Analytical Framework

The simulated dataset was analyzed using PostgreSQL queries.

Key analytical metrics include:

Total Premium Collected (2024)

₹223.9 Million Represents total revenue generated from all policies sold during 2024.

Monthly Claim Cost (2025–2026)

Claims were aggregated monthly to analyze claim patterns.

Findings show:

• Monthly claims in 2025 ranged between **₹31M – ₹33M**

• Claims decline in early 2026 due to a **shorter eligibility window**

Claim Ratio by Policy Tenure

Claim ratio analysis helps evaluate **profitability across policy segments**.

Results indicate:

• **1-year and 4-year policies show higher claim exposure**

• **2-year and 3-year policies demonstrate relatively balanced risk**

Future Claim Liability

Assuming every vehicle without a claim eventually files one:

Estimated Liability: ₹9.57 Billion

This highlights the **long-term financial exposure** associated with active policies.

Earned Premium Analysis

Premium revenue is recognized daily based on policy duration.

Earned Premium (as of Feb 28, 2026): ₹65.9 Million

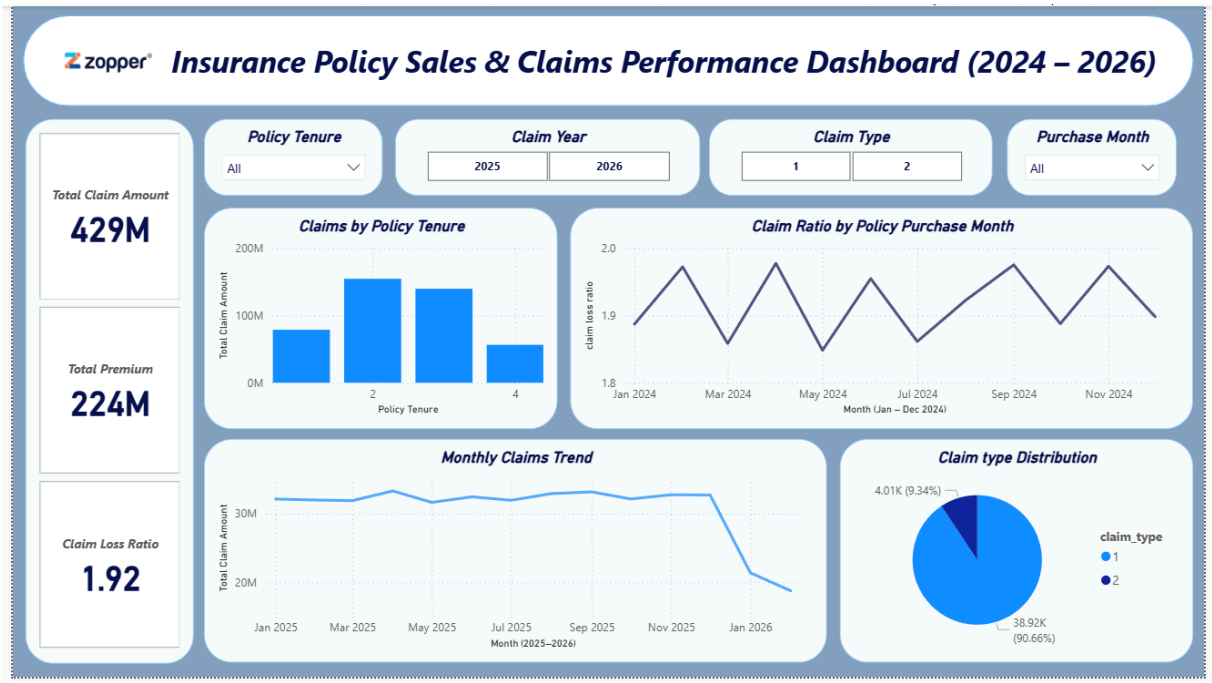
The remaining premium represents **future revenue yet to be recognized**.

4. Dashboard Overview

To enhance business interpretation, the analysis results were visualized through an **interactive Power BI dashboard titled:**

Insurance Policy Sales & Claims Performance Dashboard (2024–2026)

The dashboard provides a consolidated view of portfolio performance.



Key Performance Indicators

Metric	Value
Total Premium	₹224M
Total Claim Amount	₹429M
Claim Loss Ratio	1.92

These KPIs summarize overall portfolio profitability.

Dashboard Visualizations

- **Claims by Policy Tenure**
  Shows how claim costs vary across tenure segments.

- **Claim Ratio by Policy Purchase Month**
  Illustrates the relationship between policy purchase timing and claim exposure.

- **Monthly Claims Trend**

Tracks claim activity from **2025 to early 2026**, highlighting stable claim patterns followed by a decline in early 2026.

- **Claim Type Distribution**
  Displays the proportion of **first claims vs repeat claims**, providing insight into claim frequency patterns.

Interactive Filters

Users can dynamically analyze the data using slicers for:
• Policy Tenure
• Claim Year
• Claim Type
• Purchase Month
These filters enable **segmented analysis of claim patterns and premium performance**.

5. Key Insights

- ### Premium Contribution
  Policies with longer tenure generate higher premium revenue.
  3-year policies contribute the largest share of premium income due to their higher portfolio share.

- ### Claim Stability
  Monthly claim values during 2025 remain relatively stable, indicating **consistent claim occurrence patterns** in the simulated portfolio.

- ### Balanced Risk Across Purchase Months
  Claim ratios remain relatively consistent across policy purchase months, suggesting **limited seasonal impact on claims**.

- ### Long-Term Portfolio Risk
  The estimated **₹9.57B potential claim liability** highlights the importance of financial reserves and actuarial forecasting in insurance operations.

- **Earned vs Unearned Premium**
  As of February 2026, a portion of premium remains unearned due to multi-year policies still active.
  This represents **future revenue potential but also continued claim exposure**.

6. Conclusion

This analysis demonstrates how simulated insurance datasets can be used to evaluate **portfolio performance, claim behavior, and financial risk exposure**.
By combining **SQL-based data simulation, analytical queries, and Power BI visualization**, the project provides a comprehensive view of insurance portfolio dynamics.
The approach highlights the importance of **data-driven decision making in pricing strategy, risk assessment, and portfolio management**, which are essential functions within modern **Business Intelligence and insurance analytics teams**.